```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("restaurant_customer_data_600.csv")
print(df.head())
```

```
   CustomerID  Age  Gender  VisitFrequency  AvgBill  TimeSpent  Rating  \
0        1001   56  Female              13     1368         67     4.5
1        1002   46    Male               4      415         56     4.9
2        1003   32  Female              12      202         34     3.4
3        1004   25    Male              13      359         54     4.1
4        1005   38    Male               2      980        111     5.0

    FavCategory     FavDish
0        Drinks  Lemon Soda
1  South Indian        Dosa
2     Desserts     Brownie
3        Drinks  Lemon Soda
4     Fast Food       Fries
```

```
print(df.tail())
```

```
     CustomerID  Age  Gender  VisitFrequency  AvgBill  TimeSpent  Rating  \
595        1596   49  Female              10     1449         34     3.2
596        1597   38    Male               3      872         89     4.7
597        1598   33    Male               2      830         66     3.5
598        1599   38  Female               2      860         97     4.0
599        1600   28    Male              14      861         94     3.3

      FavCategory       FavDish
595        Drinks        Mojito
596  South Indian          Idli
597       Chinese    Manchurian
598         Pizza  Peppy Paneer
599         Pizza  Peppy Paneer
```

```
print(df)
```

```
     CustomerID  Age  Gender  VisitFrequency  AvgBill  TimeSpent  Rating  \
0          1001   56  Female              13     1368         67     4.5
1          1002   46    Male               4      415         56     4.9
2          1003   32  Female              12      202         34     3.4
3          1004   25    Male              13      359         54     4.1
4          1005   38    Male               2      980        111     5.0
..          ...  ...     ...             ...      ...        ...     ...
595        1596   49  Female              10     1449         34     3.2
596        1597   38    Male               3      872         89     4.7
597        1598   33    Male               2      830         66     3.5
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      FavCategory       FavDish
0          Drinks    Lemon Soda
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2        Desserts       Brownie
3          Drinks    Lemon Soda
4       Fast Food         Fries
..            ...           ...
595        Drinks        Mojito
596  South Indian          Idli
597       Chinese    Manchurian
598         Pizza  Peppy Paneer
599         Pizza  Peppy Paneer

[600 rows x 9 columns]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   CustomerID      600 non-null    int64
 1   Age             600 non-null    int64
```

```
 2   Gender          600 non-null    object
 3   VisitFrequency  600 non-null    int64
 4   AvgBill         600 non-null    int64
 5   TimeSpent       600 non-null    int64
 6   Rating          600 non-null    float64
 7   FavCategory     600 non-null    object
 8   FavDish         600 non-null    object
dtypes: float64(1), int64(5), object(3)
memory usage: 42.3+ KB
```

```
df.describe()
```

	CustomerID	Age	VisitFrequency	AvgBill	TimeSpent	Rating
count	600.000000	600.000000	600.000000	600.000000	600.000000	600.000000
mean	1300.500000	39.193333	7.413333	879.525000	66.511667	3.996667
std	173.349358	12.138844	4.044307	381.552841	29.559818	0.591472
min	1001.000000	18.000000	1.000000	201.000000	20.000000	3.000000
25%	1150.750000	29.000000	4.000000	551.250000	40.000000	3.500000
50%	1300.500000	40.000000	7.000000	903.500000	64.000000	4.000000
75%	1450.250000	50.000000	11.000000	1201.000000	92.000000	4.500000
max	1600.000000	59.000000	14.000000	1499.000000	119.000000	5.000000

```
df.describe(include='object')
```

	Gender	FavCategory	FavDish
count	600	600	600
unique	2	6	18
top	Male	Pizza	Brownie
freq	310	110	45

```
df.shape
```

```
(600, 9)
```

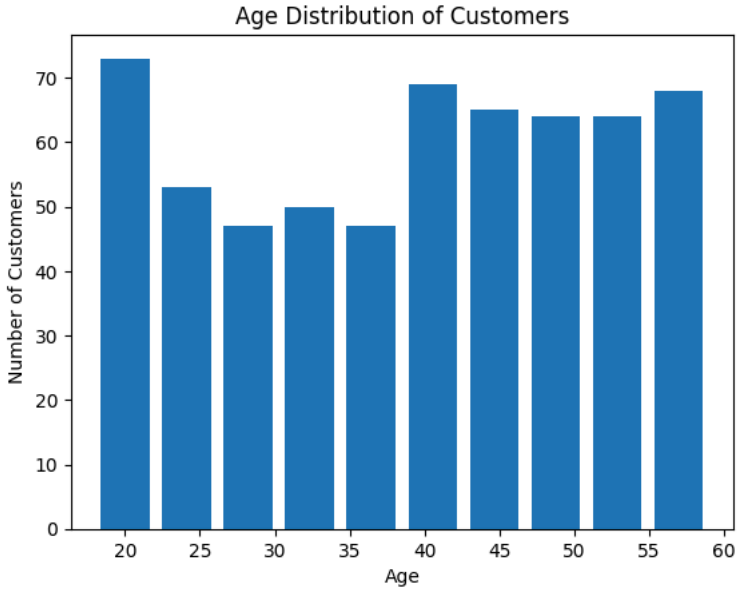
```
df.isnull().sum()
```

	0
CustomerID	0
Age	0
Gender	0
VisitFrequency	0
AvgBill	0
TimeSpent	0
Rating	0
FavCategory	0
FavDish	0

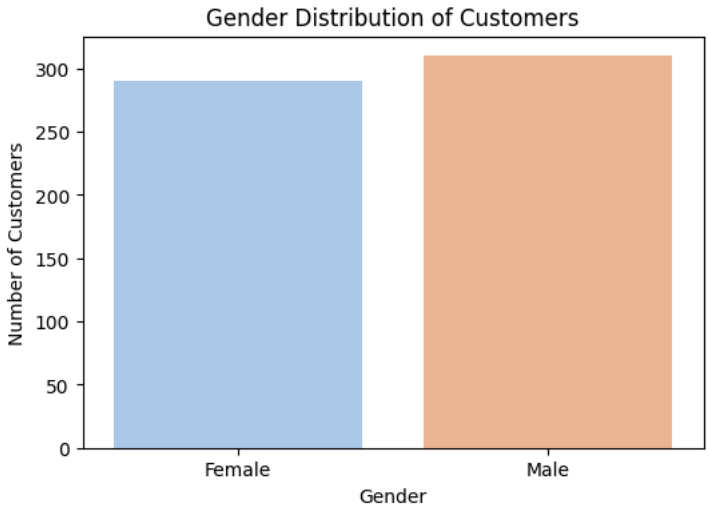
dtype: int64

```
import matplotlib.pyplot as plt

plt.hist(df['Age'], bins=10, rwidth=0.8)

plt.title("Age Distribution of Customers")
plt.xlabel("Age")
plt.ylabel("Number of Customers")
```

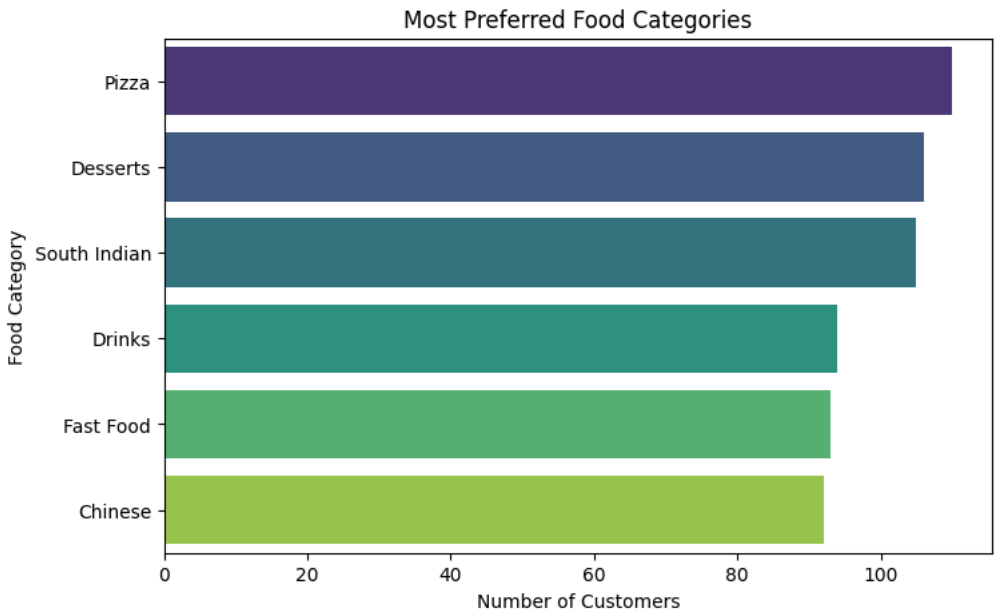
```
plt.show()
```



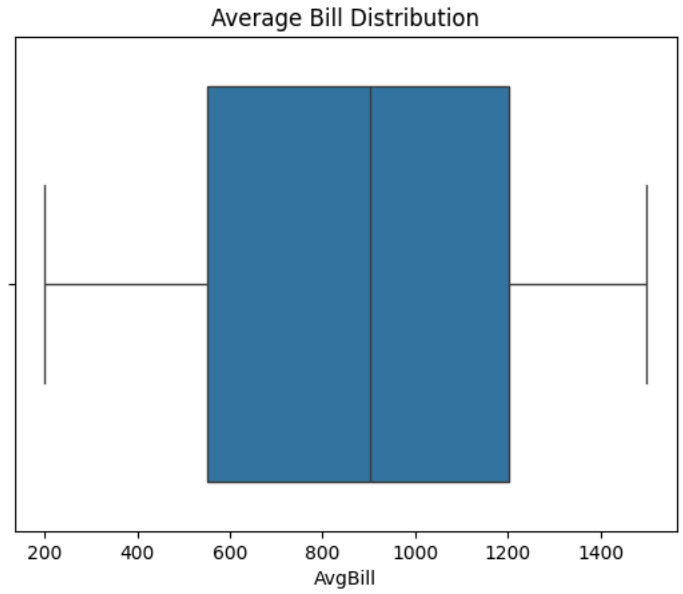

```
plt.figure(figsize=(6,4))

sns.countplot(x='Gender', data=df, hue='Gender', palette='pastel', legend=False)

plt.title("Gender Distribution of Customers")
plt.xlabel("Gender")
plt.ylabel("Number of Customers")

plt.show()
```




```
import warnings
warnings.filterwarnings("ignore")
category_counts = df['FavCategory'].value_counts()

plt.figure(figsize=(8,5))

sns.barplot(x=category_counts.values, y=category_counts.index, palette='viridis')

plt.title("Most Preferred Food Categories")
plt.xlabel("Number of Customers")
plt.ylabel("Food Category")

plt.show()
```

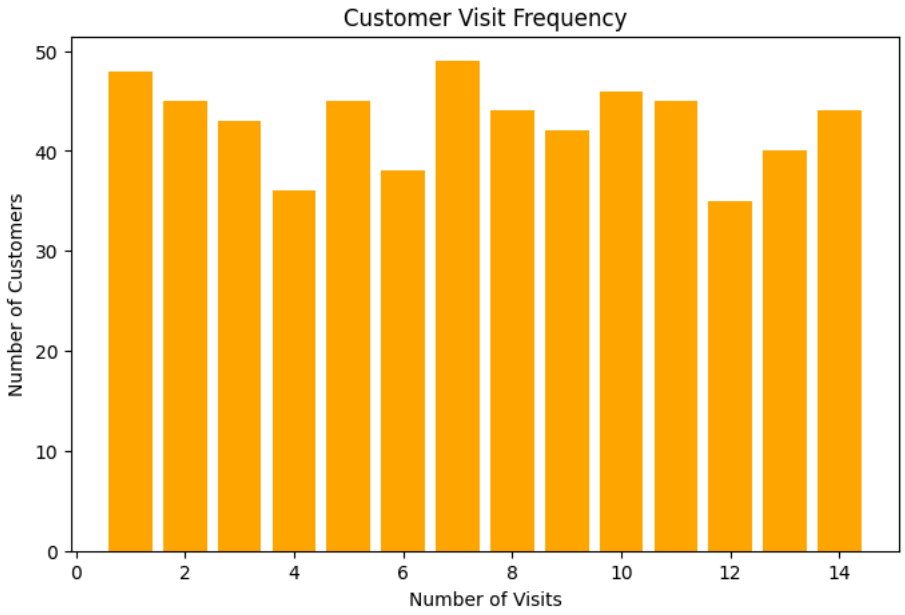



```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x=df['AvgBill'])

plt.title("Average Bill Distribution")

plt.show()
```

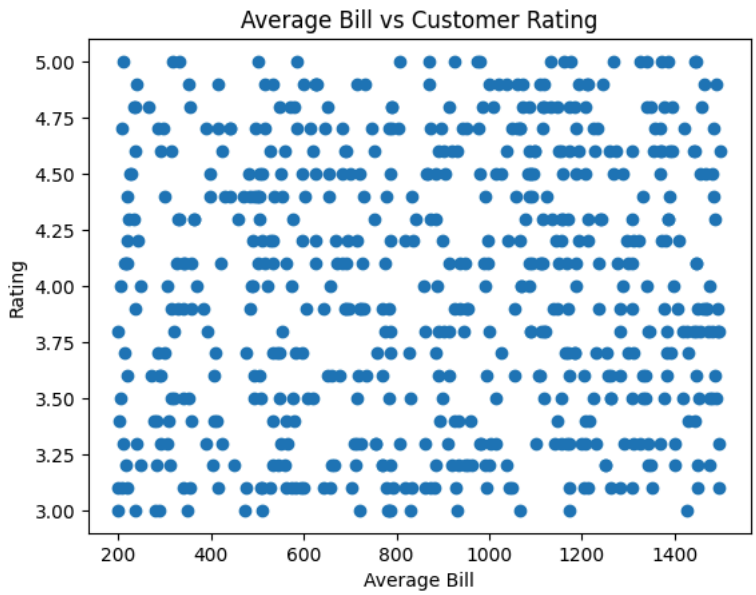



```
plt.figure(figsize=(8,5))

sns.countplot(x='Rating', data=df, palette='coolwarm')

plt.title("Customer Rating Distribution")
plt.xlabel("Rating")
plt.ylabel("Number of Customers")

plt.show()
```

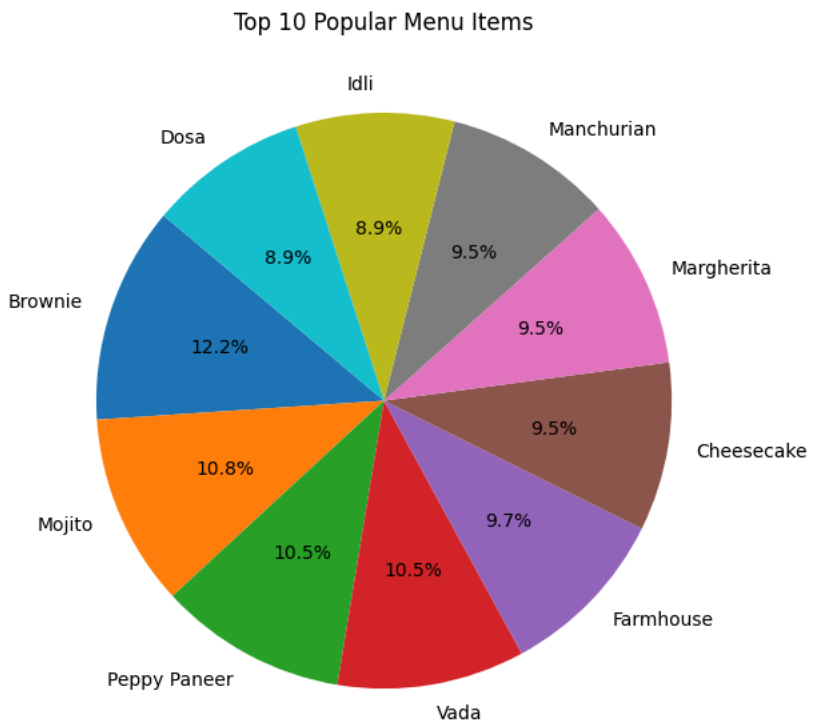
```
visit_counts = df['VisitFrequency'].value_counts().sort_index()

plt.figure(figsize=(8,5))

plt.bar(visit_counts.index, visit_counts.values, color='orange')

plt.title("Customer Visit Frequency")
plt.xlabel("Number of Visits")
plt.ylabel("Number of Customers")

plt.show()
```



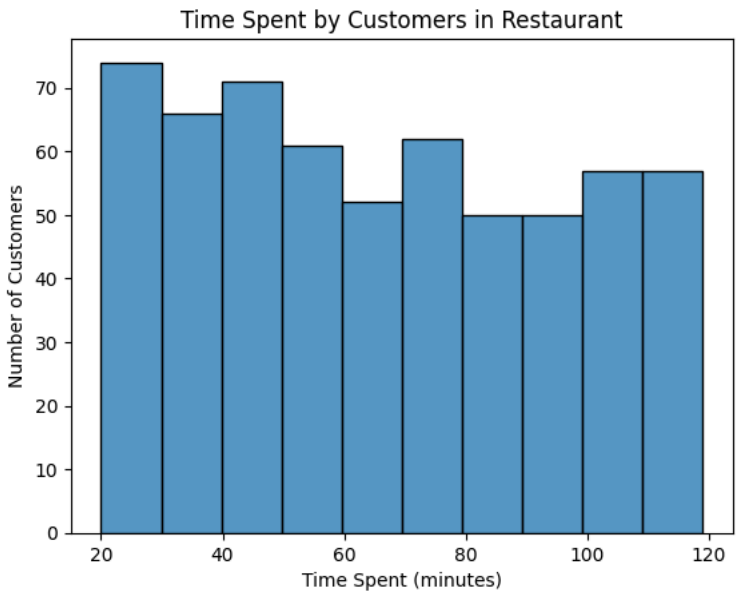

```
plt.scatter(df['AvgBill'], df['Rating'])

plt.title("Average Bill vs Customer Rating")
plt.xlabel("Average Bill")
plt.ylabel("Rating")

plt.show()
```




```
top_dishes = df['FavDish'].value_counts().head(10)

plt.figure(figsize=(7,7))

plt.pie(top_dishes.values, labels=top_dishes.index, autopct='%1.1f%%', startangle=140)

plt.title("Top 10 Popular Menu Items")

plt.show()
```




```
sns.histplot(df['TimeSpent'], bins=10)

plt.title("Time Spent by Customers in Restaurant")
plt.xlabel("Time Spent (minutes)")
plt.ylabel("Number of Customers")

plt.show()
```



Start coding or generate with AI.